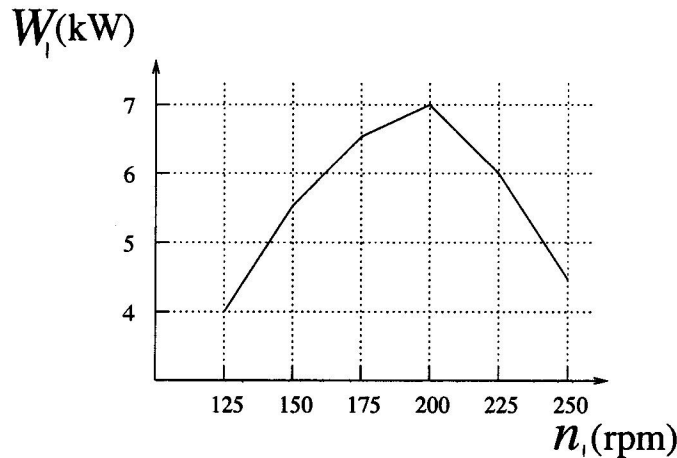


We want to investigate the performance of a wind turbine of given geometrical shape and diameter D rotating with angular velocity n subject to a wind stream of velocity U . Assume in the development that the velocity is sufficiently small compared with that of sound that air effectively behaves as an incompressible fluid of constant density and kinematic viscosity ρ and ν .

- Use the dimensional analysis to reduce the parametric dependence of the power produced by the wind turbine W .
- Simplify the results by assuming that the effect of viscosity μ is negligible. Justify the simplification.

We have measured the variation with n of the power delivered by a given turbine of diameter $D = 5$ m on a day when the wind was blowing at $U = 10$ m/s, given the results shown in the plot below.



Based on these results, address the following three questions:

- If the wind velocity on a different day is $U = 15$ m/s, determine the angular velocity required for the turbine to deliver maximum power as well as the resulting power.
- If we fix the angular velocity at $n = 200$ rpm, obtain the power response as a function of the wind velocity $W(U)$.
- Consider a geometrically similar turbine of diameter $D = 10$. Obtain the power that it would deliver if placed rotating at $n = 175$ rpm in a wind stream of velocity $U = 20$ m/s.

$$W = W(U, n, D, \rho, \mu) \rightarrow \frac{W}{\rho U^3 D^2} = f\left(\frac{n}{U/D}, \frac{\rho U D}{\mu}\right)$$

- TO BE AT THE MAXIMUM CONDITIONS HAVE TO BE PHYSICALLY SIMILAR TO THE MAXIMUM IN THE FIGURE

$$\frac{n_1}{U_1/D} = \frac{n_2}{U_2/D} \rightarrow n_2 = \frac{U_2}{U_1} n_1 = 300 \text{ rpm} \rightarrow \frac{W_2}{\rho U_2^3 D^2} = \frac{W_1}{\rho U_1^3 D^2} \rightarrow W_2 = \left(\frac{U_2}{U_1}\right)^3 W_1 = 23.625 \text{ kW}$$

- IN TYPICAL APPLICATIONS $\frac{\rho U D}{\mu} \gg 1$ SO THAT VISCOSITY DOES NOT INFLUENCE THE FLUID MOTION SIGNIFICANTLY

$$\frac{W}{\rho U^3 D^2} = f\left(\frac{n}{U/D}\right)$$

- FOR EACH n IN THE CURVE WE OBTAIN THE VELOCITY FOR WHICH CONDITIONS ARE PHYSICALLY SIMILAR ACCORDING TO $U = \frac{200}{n_1} \times 10 \text{ m/s}$ AND DETERMINE THE CORRESPONDING POWER $W = \left(\frac{U}{10}\right)^3 W_1$ GIVING THE RESULTS

U (m/s)	16	13.3	11.43	10	8.89	8
W (kW)	16.33	13.03	9.7	7	4.2	2.3

- THE POINT THAT IS PHYSICALLY SIMILAR TO THE OPERATIONAL CONDITIONS INDICATED IS $n = \left(\frac{20}{10}\right) \left(\frac{5}{10}\right) 175 \text{ rpm} = 175 \text{ rpm}$ AND THE CORRESPONDING POWER IS

$$W = \left(\frac{20}{10}\right)^3 \left(\frac{10}{5}\right)^2 6.5 \text{ kW} = 208 \text{ kW}$$